

c) $f(m,n) = m+n+1$

Solution: Let, $f(m,n) = x$, where $x \in \mathbb{Z}$.

By Defn. 1 from Sec-1.7, x can either be even or odd.

Let, x be odd. $\therefore \exists k \in \mathbb{Z}$ s.t. $x = 2k+1$ (By Defn. 1 from Sec-1.7)

If $m=n=k$, then $m+n+1 = k+k+1 = 2k+1 = x$.

Let, x be even. $\therefore \exists k \in \mathbb{Z}$ s.t. $x = 2k$ (By Defn. 1 from Sec-1.7)

If $m=(k-1)$ and $n=k$ then $m+n+1 = (k-1)+k+1 = 2k = x$.

$k-1 \in \mathbb{Z}$, this is easy to prove.

$\therefore \forall x \in \mathbb{Z}, \exists m,n \in \mathbb{Z}$ s.t. $f(m,n) = x$. By Defn. 7, $f(m,n)$ is onto Π

d) $f(m,n) = |m| - |n|$

Solution: Let, $f(m,n) = x$, where $x \in \mathbb{Z}$.

If, $x \geq 0$, then let $m=2x$ and $n=x$.

$$|m| - |n| = |2x| - |x| = 2x - x = x$$

If $x < 0$, then let $m=x$ and $n=2x$

$$|m| - |n| = |x| - |2x| = -x - (-2x) = x$$

We know that $\because x \in \mathbb{Z}, 2x \in \mathbb{Z}$.

$\therefore \forall x \in \mathbb{Z}, \exists m,n \in \mathbb{Z}$ s.t. $f(m,n) = x$. By Defn. 7, $f(m,n)$ is onto Π

e) $f(m,n) = m^2 - 4$

Solution: Let $f(m,n) = 1$. We know $1 \in \mathbb{Z}$.

$$\therefore m^2 - 4 = 1 \Rightarrow m^2 = 5 \Rightarrow m = \pm\sqrt{5} \notin \mathbb{Z}.$$

By Defn. 7, $f(m,n)$ is not onto Π

b) Determine whether the function $f: \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ is onto if

a) $f(m,n) = m+n$.

Solution: Let, $f(m,n) = x$, where $x \in \mathbb{Z}$. $\therefore m+n = x$.

If $m=x$, then $n=0 \in \mathbb{Z}$.

$\therefore \forall x \in \mathbb{Z}, \exists m,n \in \mathbb{Z}$ s.t. $f(m,n) = x$. By Defn. 7, $f(m,n)$ is onto Π